Param Mehta

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EDUCATION

University of Calgary — **Calgary, AB**
Bachelor of Science, Computer Science — Sept. 2026 – 2030 (expected)

Sir Winston Churchill High School — **Calgary, AB**
High School Diploma — Honours with Distinction; IB Computer Science HL — Sept. 2023 – June 2026
Certifications: Codeyoung — Java (Levels 1–4), Python (Levels 1–2), Machine Learning. Honour Roll 2023–2026.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS

Libraries & Frameworks: FastAPI, Pydantic, sentence-transformers (Hugging Face), NumPy, scikit-learn, SciPy, NetworkX, Shapely, PyProj, scikit-image, Spring Boot, Spring Security, Spring Data JPA, Swing

Tools & Data: Docker, Locust, Uvicorn, Git/GitHub, Maven, REST APIs & OpenAPI, SQLite, H2, GeoJSON, QGIS, OpenStreetMap/Overpass, ArcGIS REST, coordinate reference systems (EPSG:3780, WGS84)

PROJECTS

UCalgary Wayfinder — Campus Routing Engine — **Sept. 2026**
Python, NetworkX, Shapely, PyProj, SciPy, JavaScript · AI-assisted development (prompt engineering)

- Built the pipeline that turns room polygons extracted from 518 Archibus floor-plan PDFs into one routable campus graph — **87.5K nodes, 95.1K edges, 495 floors, 76 buildings** — stacking floors on detected vertical cores and linking buildings over OpenStreetMap ways.
- Georeferenced each floor onto the university's ArcGIS footprints by maximising IoU (rotation per building, scale and position per floor) for **0.95 median coverage**; solved in metres (EPSG:3780), exported as WGS84 GeoJSON.
- Derived doors from shared wall boundaries and paths from corridor centrelines so routes pass through real doorways rather than walls; automated verification reports **90.3% room reachability** and vertical alignment on every build.
- Shipped a zero-dependency viewer running client-side Dijkstra over a 2 MB gzipped graph, with a step-free mode that routes around stairs.
- Directed the build through an AI coding agent via iterative prompt engineering — specified the data model and algorithms, diagnosed real pipeline failures (CAD-layer parsing, georeferencing drift, broken corridor topology) directly from tool output, and iterated to the verification numbers above.

Vector Search & Recommendation Microservice — **2026**
Python, FastAPI, sentence-transformers, NumPy, Pydantic, Docker, Locust

- Built a FastAPI service that embeds a product catalogue into 384-dimensional vectors with the all-MiniLM-L6-v2 sentence transformer, serving semantic search over Pydantic-validated JSON endpoints.
- Took model inference off the request path by pre-computing embeddings into a cached 5,000 × 384 NumPy matrix, cutting a query to one vectorised cosine-similarity pass — **22 ms average latency**, 350 MB resident.
- Load-tested with Locust over 1,000 concurrent requests: **41 ms p95 latency** and ~265 queries/second on a single Uvicorn worker.
- Packaged the runtime, dependencies and vector index into a Docker image, with OpenAPI/Swagger docs generated from the endpoint schemas.

Currency Converter API — **2026**
Java, Spring Boot, Spring Data JPA, Spring Security, H2, Maven

- Built a Spring Boot REST service exposing /convert, /currencies and /history, persisting each conversion to an H2 database via Spring Data JPA and gating the endpoints behind a custom API-key filter in the Spring Security chain.

Haven — Rental Management System — **2025–2026**
Java, Swing, SQLite, JDBC — IB Computer Science HL Internal Assessment

- Built a two-role desktop app (landlord and renter dashboards) over a SQLite schema of users, properties, messages and agreements; carried it through the full IA cycle from client requirements and UML design to evaluation.

EXPERIENCE

Treasury Ambassador — **Nov. 2025 – July 2026**
Youth Leaders in Law — Calgary, AB

- Researched and pitched grant proposals to secure funding for the organisation's initiatives and events.
- Ran segments of the annual YLL Conference at the University of Calgary Faculty of Law — moderated a mock trial, led campus tours, managed a corporate-law panel Q&A.

Hackathon Winner — 1st Place Overall — **Dec. 2024**
Amazon Web Services (AWS) TechnoTrailblazers — Calgary, AB

- Placed first overall in AWS's TechnoTrailblazers hackathon, run at selected high schools across Canada; built a sustainability-focused solution in Java under a fixed time limit, contributing across ideation, feasibility analysis, solution design, and implementation.

LEADERSHIP & AWARDS

Competitive Debater, Sir Winston Churchill High School (2023–2026) — Top-4 team finishes at three tournaments including the University of Calgary's Alithea; 6th place speaker at Alberta Regionals. Chief Adjudication Personnel at two tournaments.

Model United Nations, SWCHSMUN (2025–2026) — Most Outstanding Delegate (WISEMUN); Most Improved Delegate (SWCHSMUN).